Case Studies in Thermal Engineering 19 (2020) 100607

Contents lists available at ScienceDirect

Case Studies in Thermal Engineering

journal homepage: http://www.elsevier.com/locate/csite

An innovative technique of passive cooling PV module using lapping fins and planner reflector

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ARTICLE INFO

Keywords:
Passive cooling
Efficiency enhancement
Experimental investigation
Computational fluid dynamics (CFD)
EES

ABSTRACT

This study aims to combine planar reflector and backplate extended surface to the PV module to optimize its efficiency. A parametric study based on lapping fin height, number fins, fin spacing, windspeed and solar irradiance on the PV module performance were investigated using CFD, ANSYS, FLUENT. 3-D, Navier–Stokes energy equations were solved using ANSYS, FLUENT to perform numerical computations. Meanwhile, the bare PV module temperature has been calculated using Engineering Equation Solver (EES) software. The results revealed that as the fin pitch increases from 20 to 60 mm this leads to decrease number of fins from 20 to 10 fins which at the same time increased the PV module temperature from 44.13-54.01 °C. The PV module temperature without cooling was reached 64.3 °C; meanwhile with 18 lapping fins and 27.7 mm fin pitch the PV module temperature reduced to 39.73 °C which improved the electrical efficiency to 11.2% when compared to bare PV 9.81% with a temperature difference 24.57 °C at 1000 W/m^2. Finally, we conclude that the lapping fins have a superior performance in reducing PV module temperature. Furthermore, it was found that the aluminum thickness of the designed fins above 2 mm showed a negligible effect on the PV power and efficiency.

1. Introduction

One of the weak points of PV module is the negative effects of increasing the cell's temperature during power generation [1]. The increasing in PV module temperature can be partially avoided by extracting the heat using a suitable cooling system with using different fins configuration. Where, a number of cooling methods have been attempted and reported in the previous literature. The proper cooling of PV module reduces the performance degradation and increase the lifetime of the PV module. Many researchers prefer PV module passive cooling technique as it does not require any additional power to operate [2]. In the case of passive cooling methodologies there are a lot of studies have been done. A multi-concept of cooling technique which involves three different types of cooling methods, namely, air, water, conductive passive cooling were investigated by Ref. [3]. In their experiment, two identical PV module of 250 W were mounted at 37 cm from the ground level in order to create room for air cooling. The results obtained revealed that the increase in the output power of 20.96 W with efficiency enhancement no less than 3% [1]. examined in door a simple and economic method to improve the PV module efficiency and reduce it temperature using Al- fins underneath the back surface of the PV module. Two different solar irradiance 420 W/m^2 and 620 W/m^2 were tested and the PV temperature reduction of 7.4 °C was observed, this leads to enhance the efficiency by 2.72% [4]. presented a novel cooling system based on evaporating cooling concept with using pin fins as heat sink. The results revealed that the PV efficiency and the PV output power improved to 31.5% and 32.7%,

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https://doi.org/10.1016/j.csite.2020.100607
Received 23 January 2020; Received in revised form 11 February 2020; Accepted 17 February 2020
Available online 26 February 2020
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respectively, with PV module temperature reduction could be reached to 20.05%. A simulation study of a passive cooling system attached with fins using a computational fluid dynamics (CFD) was developed by Ref. [5]. The simulation results showed that the PV module surface temperature and electrical efficiency without cooling were found to be 62.78 °C and 13.24%, respectively. Meanwhile, the PV module temperature reduction was 15.13 °C with using fins [6]. proposed a passive cooling method of PV module in a finned container heat sink. In their study, a Palm wax was used as phase change material due to much cheaper among other commercial PCMs. They designed and investigated three different phase change material containers, such as tubed, grooved and finned container. The author's found that the finned container could be reduce the PV module temperature from 57.9 °C to 51.8 °C with temperature difference about 6.1 °C in which the finned container exhibited the best cooling technique when compared to others. This temperature reduction leads to improve the PV module efficiency from 9.33% to 9.82% [7]. presented an innovation based on natural convection through drilled a holes in the PV module to minimize the PV temperature and enhance its performance. The drilled and the conventional PV modules are simulated using Computational Fluid Dynamics package, ANSYS. Fluent. The authors reported that the perforated PV module temperature with 6 holes decreased from 70 °C to 54 °C with temperature difference 16 °C [8]. carried out another passive cooling method with using Al-fins attached underneath the Si-poly, 50 W PV module with an epoxy conductive glue. The experimental results showed that the perforated and positioned fins enhanced the efficiency by 2% compared to convention PV module [9]. developed a passive thermal regulation technique with fins in conjunction with cotton wicks for cooling flat mc-Si PV module (25WP). The average ambient temperature was varied from 30-37 °C. The results indicated that the maximum PV module temperature at 12 p.m. without and with cooling was 49.2 °C and about 43.3 °C respectively achieving 5.9 °C temperature reduction [10]. experimentally investigated a simple passive low-cost strategy using an aluminum heat sink system attached underneath the PV module. The authors reported that the mean cell temperatures for the three lengths of heat sink fins were found 60.9 °C, 56.6 °C and 52.7 °C, respectively. It was found that the longest fin dissipates the largest amount of heat, leading to a temperature decrease of 8.2 °C with respect to the shortest fin [11]. investigated the effect of using a thermal grease between the Al-heat sink and the rear surface of the PV cell in order to improve the thermal conductivity. Under three implemented values of solar irradiance (400, 600 and 800 W/m^2); the results showed that the amount of heat dissipated from the PV cell with and without fins were (5.49, 9.06 and 12.03 W) and (4.54, 6.91 and 9.61 W), respectively. An experimental study of a single concentrator solar cell (using lens) was carried out by Ref. [12]. The thermal performance was investigated based on number of fins, fin height, fin thickness and fin spacing. The results revealed that the PV module temperature decreased with increasing of number of fins or base plate thickness.

In this paper, the electrical power and the efficiency of the PV module with using Lapping fins-concept planner reflector as an innovative passive cooling technology to help to decrease the PV module operating temperature was simulated. A 3-D, Navier–Stokes energy equations were solved using ANSYS, FLUENT, CFD software program to perform numerical computations and then experimentally validated. From other hand, a conventional PV module temperature without cooling has been calculated using Engineering Equation Solver (EES) software.

2. PV module efficiency

Mainly the PV module efficiency was affected by the ambient temperature and PV module temperature as well. The maximum power of the PV module as expressed in equation (1) [13,14]:

$$P_{mp} = V_{mp} * I_{mp} = V_{oc} * I_{sc} * FF \tag{1}$$

Where, P_{mp}, V_{mp}, I_{mp}, V_{oc} and I_{sc} stand for maximum power, voltage, current, open circuit voltage and short circuit current of the PV module, respectively, while, FF indicates to the fill factor.

As the PV module temperature rises, the short circuit current increases with a small amount, while the FF and the open circuit voltage reduce in magnitude [3]. Equation (2) shows the PV module efficiency as expressed in the form of energy output divided by energy input [15]:

$$\eta = E_{out}/E_{in} \tag{2}$$

Also, the PV module efficiency can be written as in equation (3):

$$\eta = P_m/G*A \tag{3}$$

Where, (P_m, G and A) are the maximum power, solar irradiance and the area of the PV module. The PV module efficiency can also be expressed as illustrates in equation (4) [16]:

$$\eta_{pv} = \eta_{erf\ Temp}\left[1 - \beta T_{pv} - T_{ref\ Temp}\right] \tag{4}$$

Where, η_{pv} stands for PV module efficiency, $\eta_{erf\ Temp}$ is the PV module efficiency at reference temperature ($T_{ref\ Temp} = 25$ °C at STC) and β is stands for power temperature coefficient. The temperature effect of PV module can be expressed as in equation (5) [17]:

$$P_{pv} = Y_{pv} * f_{pv}\left(G_T/G_{T,STC}\right)\left[1 + \alpha_p(T_C - T_{C,STC})\right] \tag{5}$$

Where, G_T is the incident solar radiation on the PV module (KW/m^2), f_{pv} is the derating factor (%), Y_{pv} stands for the rated capacity of

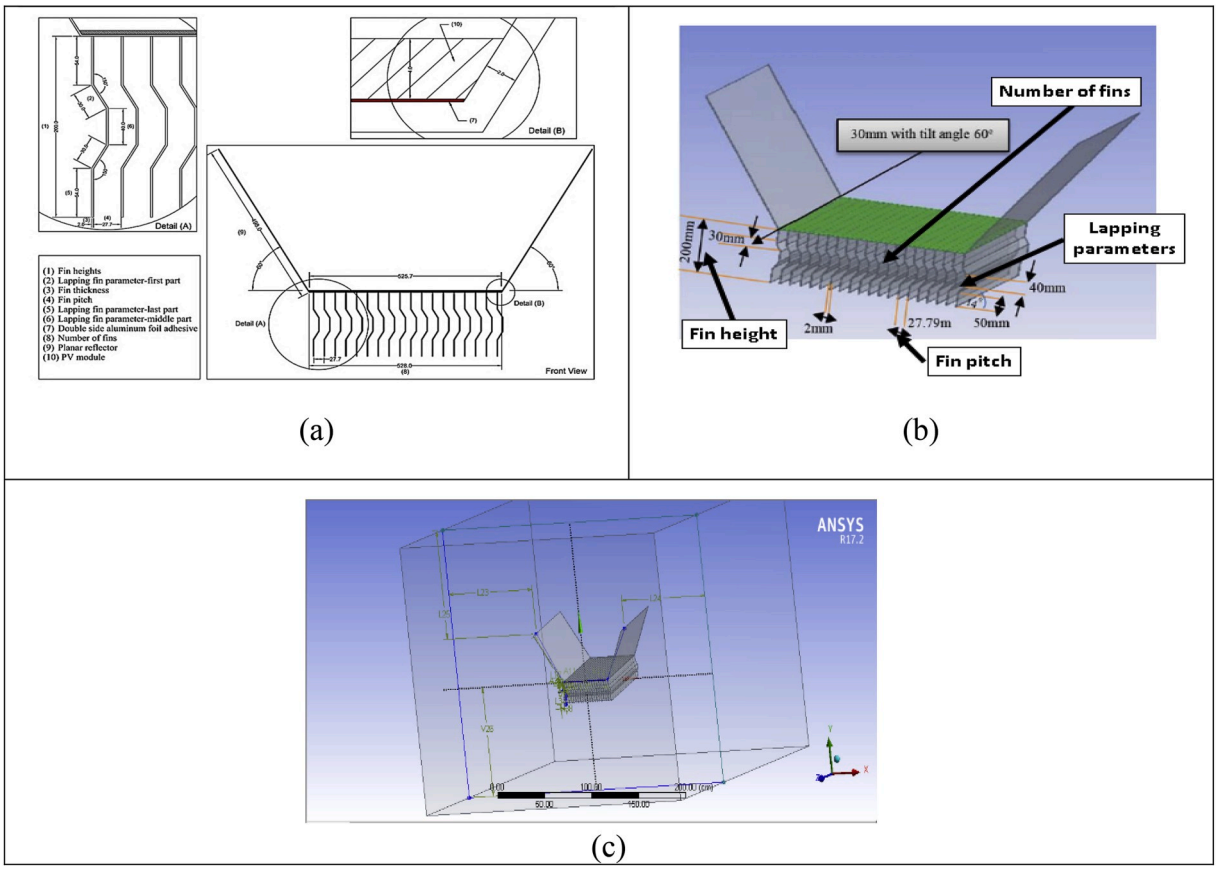


Fig. 1. (a) Dimensions of PV module with planar reflector and lapping fins, (b) geometry drew by design modeller, and (c) enclosure used as air domain.

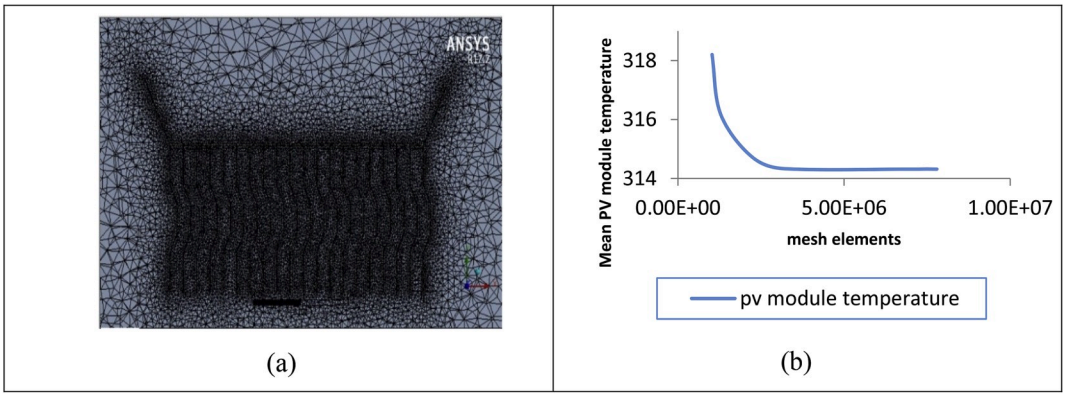


Fig. 2. (a) Geometry meshing of PV module with planar reflector and lapping fins, (b) Mean PV module temperature over mesh face for GIT.

the PV array (KW). $G_{T,STC}$ is the incident solar radiation at standard test conditions (1000 W/m^2), α_p is the power temperature coefficient, T_C is the PV cell temperature, and $T_{C,STC}$ is the PV cell temperature at (STC = 25 °C).

3. Methodology

3.1. Bare PV module temperature

The PV module surface temperature was numerically modeled and solved using Engineering Equation Solver (EES). The results showed that the electrical PV module efficiency decreased from 11.1% to 9.81% as the solar irradiance, PV module temperature and

Table 1
Lapping fin designs specifications.

Design Type	Specifications and geometry details
PV module with longitude rectangular fins	Solar irradiance, air velocity, fin height, number of fins, and tilt angle were varied from 400 – 100 Wm^{-2}, 0.5–1.5 m/s, 100–300 mm, 10–30 fins, 10–20°, whereas fin pitch for 10, 15 and 18 fins were 58.7 mm, 37.7 mm, and 27.7 mm, respectively.

Table 2
Specification of PV module with assumed working conditions.

Parameter	Unit	Value
Ambient temperature	T_a	298 K
PV length	L	670 mm
PV width	W	530 mm
Reflectors dimensions	L*W	Same dimensions of PV
Collector area	A_c	0.3551m^2
Aluminum fins thickness	L_f	2 mm
Al-Fin conductivity	k_f	204 W/m k
Aluminum plate thickness	L_{Al}	2 mm
Length of Aluminum fins	a	670 mm
PV module reference efficiency	η	12%
Air thermal conductivity	k	0.0242 W/m-k
Density of aluminum	ρ_{al}	2330 kg/m^3
Thermal conductivity of aluminum	K_{al}	148 W/m-k
Number of cells	Pcs	36 cell
Maximum power at STC	P_{max}	40 W ± 5%
Open circuit voltage at STC	V_{oc}	21.6 V
Short circuit current at STC	I_{sc}	2.57A
Voltage at Max. Power at STC	V_{mp}	17.2 V
Current at Max. Power at STC	I_{mp}	2.35A
Air specific heat	C_p	1006.43 j/kg-k
Viscosity of the air	U	1.7894e^{-5}
Specific heat of the aluminum	C_{pal}	712 j/kg-k
Air density	ρ	1.225 kg/m^3

the ambient temperature increased from (400–1000 W/m^2), (41.32°C–64.30 °C), and (28.8°C–34 °C), respectively.

3.2. CFD simulation

Passive cooling of pc-Si Solartif, 40 W PV module with planar reflector and lapping fin configuration was simulated using Computational Fluid Dynamics package, ANSYS. Fluent. Fig. 1(a) shows the Lapping fins with planar reflector and Fig. 1(b) illustrate the geometry drew by using design modeller, while Fig. 1(c) shows an enclosure used as air domain.

3.3. Grid independence test (GIT)

The GIT was performed to obtain the most suitable face size meshing for specific geometries. Fig. 2 (a), the mesh for the PV module with planar reflector and lapping fins with an inlet air velocity 1 m/s. The discretization grid was structured and uniforms. The face mesh size has been tested from 20 mm and decreased up to 5 mm. In this case, face sizing with 5 mm was chosen as the best in terms both accuracy and higher orthogonal quality as shown in Fig. 2 (b).

3.4. Specification of the PV module with lapping fins design

The specifications of the lapping fins with different parameters design is shown in Table 1 and Table 2 shows the material of the PV modules, absorber thickness, tilt angle, ambient temperature and the solar radiation, with the assumed working conditions.

4. Experimental set-up

The aim of the outdoor experimental study is to investigate the effects of different fin configurations in terms of certain parameters namely: fin length, fin pitch, fin height, tilt angle, and number of fins on the natural convection heat transfer from the considered lapping fins design. The experimental set-up consists of two identical solar-TIF, STF-040P1 PV modules, each of 40 W were installed from 1st of Mar until 30 Sep 2017 on the 7th floor at Solar Energy Research Institute (SERI), UKM,University, Bangi, Malaysia. One of the PV module used as a bare PV without cooling (Convectional PV module), while the other PV module used with adhered (18) lapping fins on aluminum plate with thickness 2 mm and then pasted by double face aluminum foil tape at the rear side of the PV

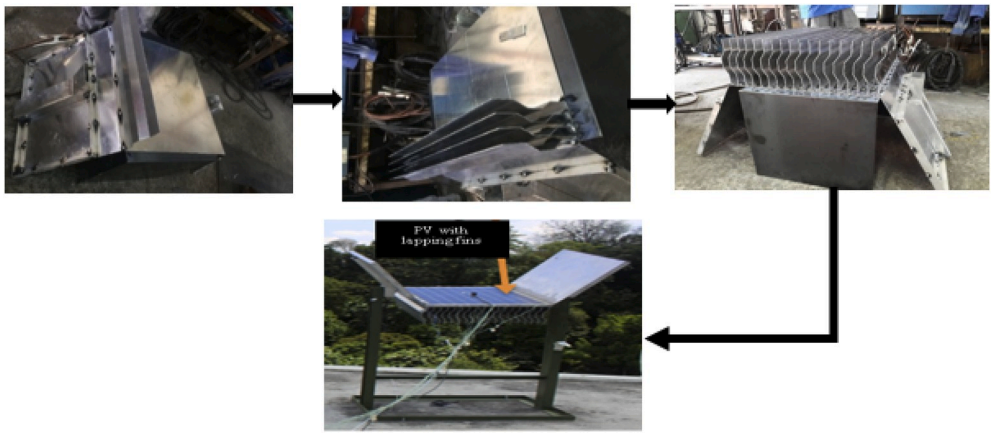


Fig. 3. Stages of designing lapping fins with planar reflector.

Table 3
Specifications of the double-sided aluminium foil tape.

Reference:	*FIRMFIX*®2418
Description:	Double-Sided Aluminium Foil Tape
Carrier:	Soft Aluminium Foil A1100:H0
Color:	Natural
Liner:	Siliconized Glassine White/Brown (Unprinted 90gsm)
Adhesive:	Synthetic Resin (Solvent Free – Low VOC)
Carrier Thickness:	50 μm/0.05 mm
Product Thickness without Liner:	180 μm/0.18 mm
Adhesive Tack/Strength:	6.5 N/cm
Tensile Strength:	20.0 N/cm
Elongation at Break:	8%
Long Term Working Temperature	80 °C
Short Term Working Temperature	160 °C
Thermal Conductivity	0.30 W/M-K
Thermal Resistance	0.78°C–In^2/W
Ageing resistance UV	Good
Humidity resistance	Excellent
Chemical resistance	Fair

module. The stages of designing lapping fins is shown in Fig. 3. Table 3 gives the specifications of the double face aluminum foil tape with higher thermal conductivity.

5. Results and discussions

5.1. CFD simulation results

Effect of varying lapping fin height from (140–200 mm), solar irradiance from (400–1000 W/m^2) and fin spacing on the PV module temperature reduction and its efficiency at different number of fins changed from (10–20) was investigated. The results revealed that the PV module temperature was decreased from 54.01 °C at 1000 W/m^2, number of fins = 10, and fin height 140 mm into 49.34 °C at 1000 W/m^2, number of fins = 10 and fin height 200 mm. While, for number of fin increases from 10 to 15 fins, the PV module temperature was decreased from 49.10 °C at 1000 W/m^2, number of fins = 15, and fin height 140 mm into 44.53 °C at 1000 W/m^2, number of fins = 15, and fin height 200 mm. Meanwhile, the PV module temperature reduced from 44.18 °C at 1000 W/m^2, number of fins = 20 and fin height 140 mm into 39.73 °C 1000 W/m^2 at number of fins = 20 with fin height 200 mm as shown in Fig. 4 (a,b). This means that the PV module temperature was significantly affected by the increasing number of fins and height of fins. Thus, for an optimum fin height 200 mm the PV module temperature reduced to about nominal cell temperature (NOCT). It can be concluded that the PV module temperature reduced from 64.3 °C without cooling to 39.73 °C with optimum fin height 200 mm and number of fins = 20 with a temperature difference about 24.57 °C. From another hand, the electrical efficiency enhanced from 9.81% without cooling into 11.2% with cooling using lapping fin. Meanwhile, the effect of fin spacing on the PV module performance enhancement at solar irradiance 1000 W/m^2, wind speed 1 m/s, and fin height 200 mm is shown in Fig. 4(C). The results revealed that as the fin pitch increases from 20 to 60 mm this leads to decrease number of fins from 20 to 10 fins which at the same time increased the PV module temperature from 44.13-54.01 °C.

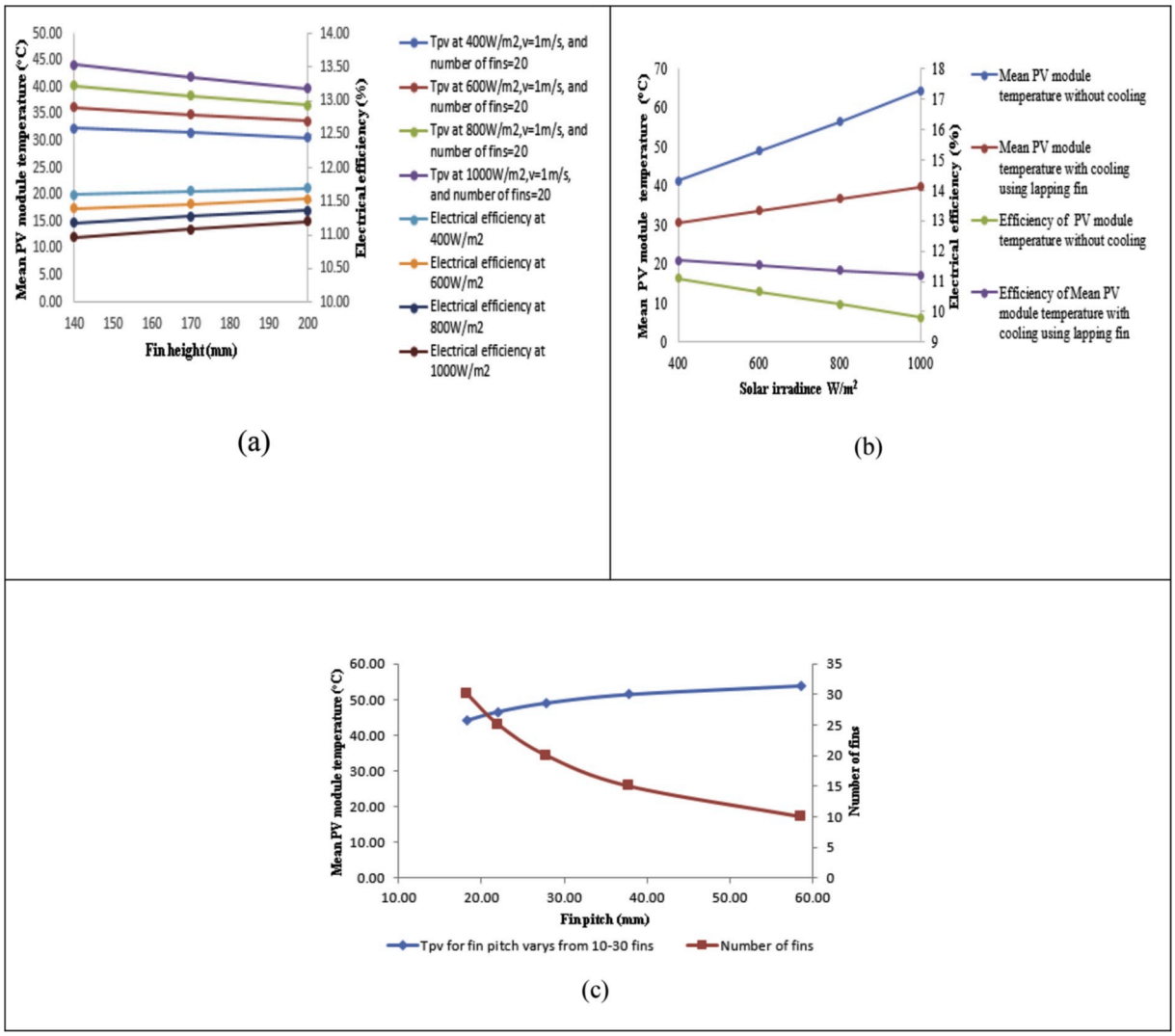


Fig. 4. Shows the effect of varying lapping fin height and solar irradiance on electrical efficiency at (a) number of fins = 20, fin height 200 mm, (b) Comparison of the mean PV module temperature reduction with lapping fin and without cooling at fin height 200 mm and number of fins = 20and (c) effect of fin spacing.

5.2. Experimental results

I–V–P curves of a bare PV model STF-040P1 PV without and with cooling is shown in Fig. 5. The output power was logged at an interval of every 1 min and the data of the current-voltage, the output power and PV module temperature were simultaneously recorded. The tilt angle 14 °C was set as an optimum tilt angle. It can be concluded that the I–V–P of the bare PV module without cooling at solar irradiance varied from 400-1000 W/m^2 resulted in changes to the short circuit current from 0.94A to 2.3A. Meanwhile, the open circuit voltage decreased from 19.6 V to 18.5 V under same conditions. Moreover, the change in the output power has been identified and was found to be increased from 11.45 W to 18.5 W. Furthermore, the fill factors (FF) reduced from 0.62 at 400 W/m^2 to 0.43 at 1000 W/m^2. In addition, the electrical efficiency decreased from 11.09% at 400 W/m^2 to 9.81% at 1000 W/m^2 as shown in Fig. 5 (a).

Meanwhile, the experimental results obtained from using PV module with planar reflector and 18 lapping fins as an optimal number of fins is shown in Fig. 5 (b). It can be concluded that for solar irradiance varied from 400-1000 W/m^2 the short circuit current, open circuit voltage, output power, and fill factor were increased from (1.1–2.65A), (19.0–20.1 V), (11.26–31.9 W), and (0.538–0.6), respectively. In addition, the validation between the simulated results obtained by CFD and the experimental results for PV module with planar reflector and 18 lapping fins compared to bare PV module temperature without cooling obtained from ESS software are shown in Fig. 5 (c). The results revealed that the numerical and the experimental results are closed to each other.

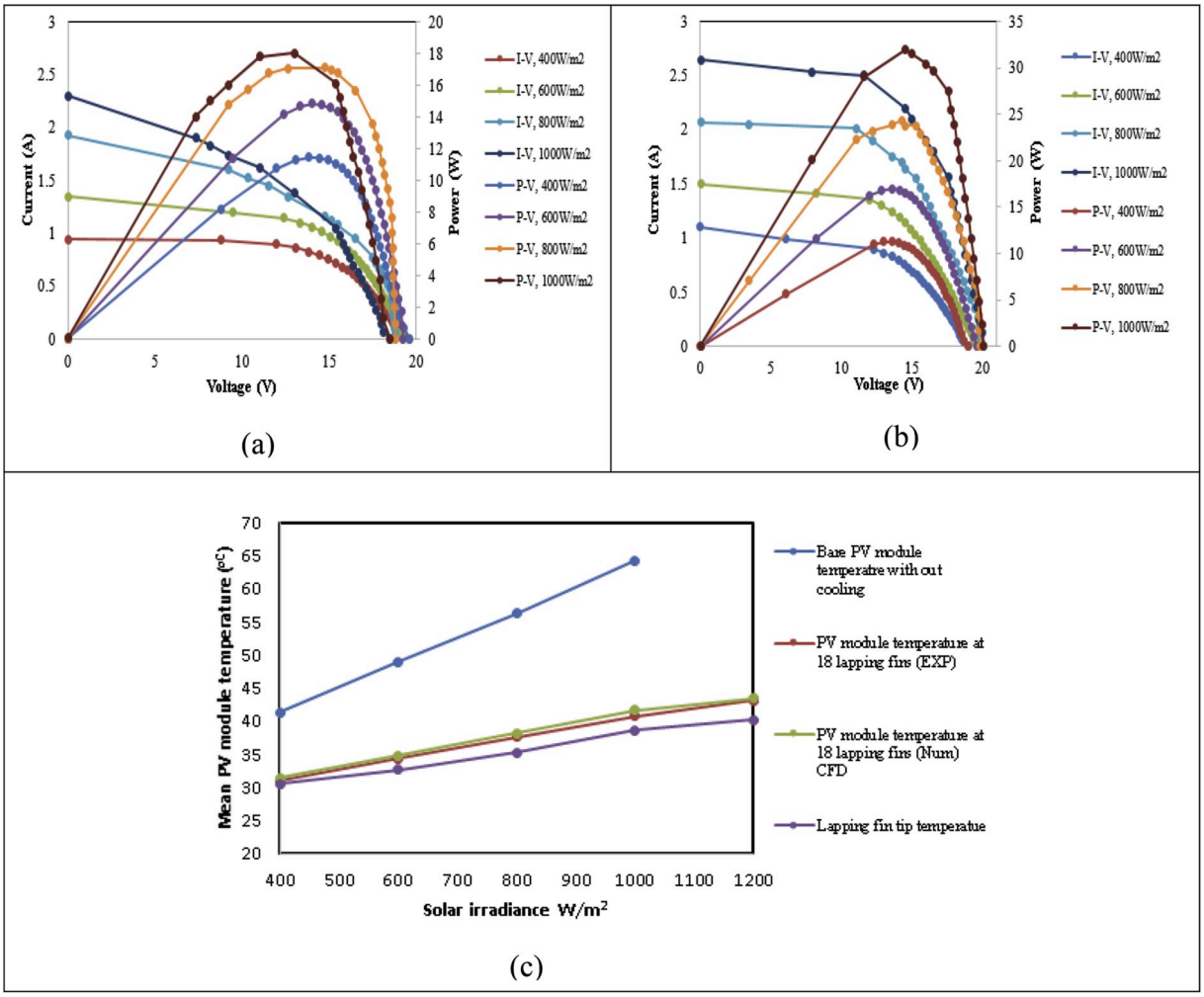


Fig. 5. Shows the (a) I–V and P–V curves for bare PV module at different solar irradiance, (b) I–V and P–V curves for PV module with reflector and (18) lapping fins and (c) comparison between the numerical simulation results & experimental results and bare PV temperature for 18 lapping fins.

6. Conclusion

The performance of PV module with 18 lapping fins was investigated and analyzed using 3-D CFD and FLUENT, ANSYS. The bare PV module temperature without fin cooling was calculated by EES and the results were increased from 41.32 °C at 400 W/m^2 to 64.3 °C at 1000 W/m^2, while the electrical efficiency decreased from 11.09% at 400 W/m^2 to 9.81% at 1000 W/m^2. Whenever, the PV module temperature reduction with planar reflector and (18) lapping fins reach up to 39.73 °C, meanwhile, the electrical efficiency improved to 11.2% compared to 9.81% without cooling. Moreover, the PV module temperature reduction with planar reflector and lapping fins was with a temperature difference about 24.57 °C when compared to reference PV module. Finally, we conclude that the lapping fins have a superior performance in reducing the PV module temperature. Furthermore, it was found that aluminum thickness of the designed fins above 2 mm showed a negligible effect on the PV power and efficiency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

A.M. Elbreki: Writing - review & editing. **K. Sopian:** Writing - original draft. **A. Fazlizan:** Investigation. **A. Ibrahim:** Supervision.

Acknowledgement

The work was performed at Solar Thermal Lab of Level 7, Universiti Kebangsaan Malaysia. The authors would like to express their appreciation to the University for providing research funding Dana Modal Insan (MI-2019-008).

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